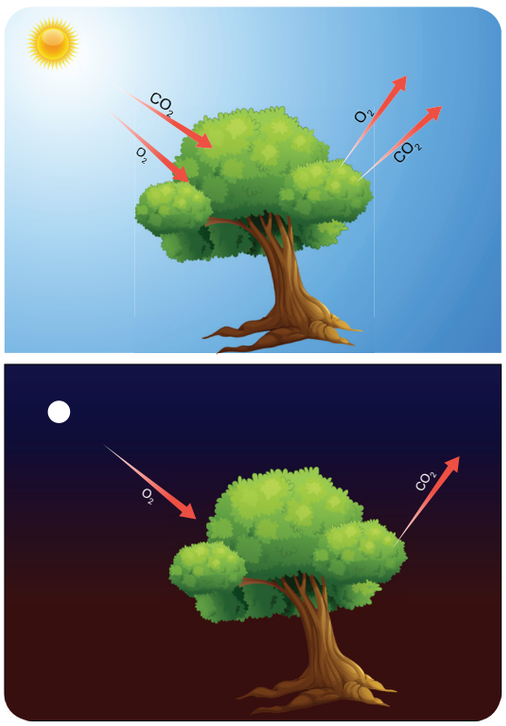


Figure 14.1: Gaseous exchange in plants

of complex organic compounds through oxidation within the cells leads to energy release. The energy released during respiration is stored in the form of **ATP** (Adenosine Tri Phosphate) as well as liberated heat. Respiration occurs in all the living cells of organisms. The overall process of respiration corresponds to a reversal of photosynthesis.

$$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{Energy}$$

(686 K cal or 2868 KJ)

(1K cal = 4.184 KJ)

Depending upon the nature of respiratory substrate, **Blackman** divided respiration into,

1. Floating respiration
2. Protoplasmic respiration

When carbohydrate or fat or organic acid serves as respiratory substrate and it is called **floating respiration.** It is a common mode of respiration and does not produce any toxic product. Whereas respiration utilizing protein as a respiratory substrate, it is called **protoplasmic respiration**. Protoplasmic respiration is rare and it depletes structural and functional proteins of protoplasm and liberates toxic ammonia.

14.1.2 Compensation point

At dawn and dusk the intensity of light is low. The point at which CO_2 released in respiration is exactly compensated by CO_2 fixed in photosynthesis that means no net gaseous exchange takes place, it is called **compensation point**. At this moment, the amount of oxygen released from photosynthesis is equal to the amount of oxygen utilized in respiration. The two common factors associated with compensation point are CO_2 and light (Figure 14.2). Based on this there are two types of compensation point. They are CO_2 compensation point and light compensation point. C_3 plants have compensation points ranging from 40-60 ppm (parts per million) CO_2 while those of C_4 plants ranges from 1-5 ppm CO_2.

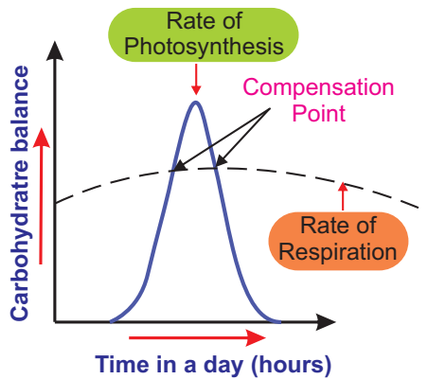


Figure 14.2: Compensation point

14.2 Structure of ATP

Respiration is responsible for generation of ATP. The discovery of ATP was made by **Karl Lohman** (1929). ATP is a nucleotide consisting of a base-adenine, a pentose sugar-ribose and three phosphate groups. Out of three phosphate groups the last two are attached by high energy rich bonds (Figure 14.3). On hydrolysis, it releases energy (7.3 K cal or 30.6 KJ/ATP) and it is found in all living cells and hence it is called **universal energy currency of the cell**. ATP is an instant source of energy within the cell. The energy contained in ATP is used in synthesis carbohydrates, proteins and lipids. The energy transformation concept was established by **Lipman** (1941).

XIth_Botany_Chapter_14_EM.indd 292 5/17/2020 9:20:58 PM